Phase 2 | Requirement Analysis

Solution Requirements

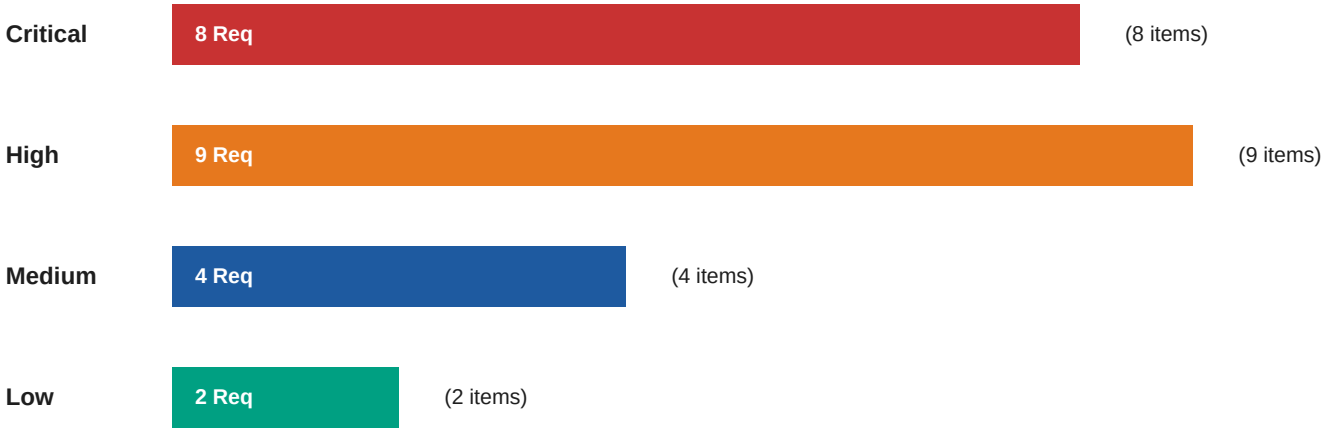
AI-Powered Travel Planner | Smart Situational Itinerary Engine

Live URL:	https://travel-planner-iota-plum.vercel.
GitHub:	github.com/itskian2007/travel-planner
Tech:	React + FastAPI + PostgreSQL + Vercel
Team:	itskian2007

1. Requirements Overview

Requirements were gathered through user research, problem statement analysis, and competitive landscape review. They are categorised as Functional (what the system does) and Non-Functional (how well it does it).

2. Requirements Priority Distribution



3. Functional Requirements

ID	Priority	Feature	Requirement Description
FR-01	CRITICAL	User Registration	System shall allow users to register with name, email & password
FR-02	CRITICAL	User Login	System shall authenticate users via email/password with JWT tokens
FR-03	CRITICAL	Forgot Password	System shall support password reset via email verification
FR-04	CRITICAL	Destination Browse	System shall display 20 global destinations with images
FR-05	CRITICAL	Itinerary Generation	System shall generate personalised 10-day itineraries
FR-06	CRITICAL	Mood Filter	Filter destinations by mood: Adventure/Relaxation/Cultural/Romance
FR-07	CRITICAL	Budget Filter	Support 3 budget tiers: Budget / Mid-Range / Luxury
FR-08	CRITICAL	Diet Filter	Support dietary: Vegetarian/Vegan/Halal/Non-Veg/Gluten-Free
FR-09	HIGH	Persona Filter	Support traveller personas: Solo/Family/Couple/Group

FR-10	HIGH	Accessibility Filter	Support accessibility preferences for itinerary customisation
FR-11	HIGH	Day 1-3 Curated	Provide expert-curated itineraries for Days 1-3
FR-12	HIGH	Day 4-10 Procedural	Auto-generate Days 4-10 via procedural engine
FR-13	HIGH	Landmark Images	Display verified landmark images via Wikipedia API
FR-14	HIGH	Destination Detail	Show full destination details: weather, culture, cuisine
FR-15	HIGH	Image Dedup	Ensure zero duplicate images across all 304 landmark images
FR-16	MEDIUM	Responsive UI	Fully responsive on mobile, tablet and desktop
FR-17	MEDIUM	Error Handling	Display user-friendly error messages for all failure states

4. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Page load time < 2 seconds on standard broadband connection
NFR-02	Performance	API response time < 500ms for itinerary generation
NFR-03	Scalability	System shall support 1000 concurrent users without degradation
NFR-04	Security	All passwords hashed with bcrypt (cost factor 12)
NFR-05	Security	JWT tokens expire after 24 hours, refresh token support
NFR-06	Security	All API endpoints protected with auth middleware
NFR-07	Availability	99.9% uptime via Vercel edge deployment
NFR-08	Usability	UI/UX follows WCAG 2.1 Level AA accessibility standards
NFR-09	Reliability	Database backups every 24 hours, point-in-time recovery
NFR-10	Maintainability	Code coverage > 80% with modular, documented components
NFR-11	Compatibility	Supports Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
NFR-12	Data Quality	All 304 landmark images verified as unique, non-duplicate